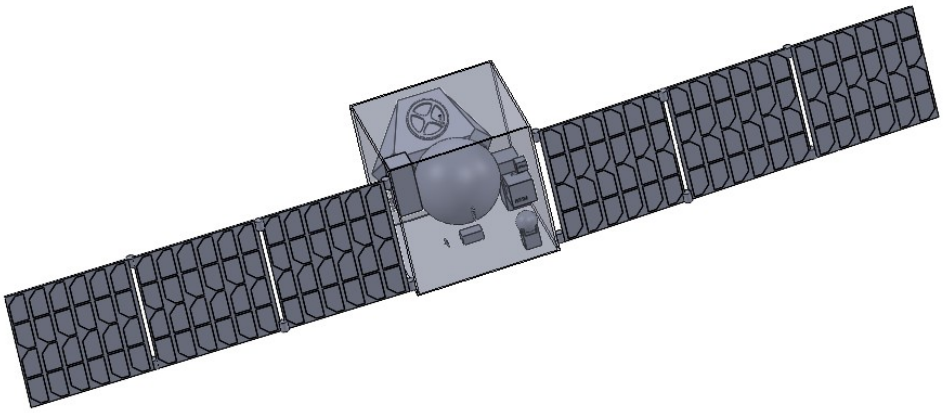
ChLOE

TECHNICAL REPORT



By

Zaara Amin: Systems Engineer

Jorge Sanz: Mission Planning & Trajectory Lead

Caden Akins: Propulsion Lead

Ignacio Garcia Perena: Power Lead

Ruelen Sarion: Structures Lead

Matthew Silva: Payload Lead, Imaging

Fotini Ioannides: Payload Lead, Refueling

Alba Borrego: Thermal Lead

Anthony Sinigiani: ACS Lead

Saahil Sinha: Telecom & C&DH Lead

Excerpt: Sections 9 and 10, written by the Power Lead. Full report by the ten-person ChLOE team, MAE 161C Spacecraft Design, UCLA, spring 2026.

9 - Power Equipment List

The Power Equipment List (PEL) catalogs the estimated power draw of every component on the spacecraft, organized by subsystem. Each component carries two power states: an active state, used when the component is operating at full capacity, and a standby state, used when the component is powered but idle. Combining these states across all subsystems produces the power demand for each of the nine operational modes defined for the mission.

Component Power Estimates

The spacecraft is divided into seven subsystems for power accounting purposes. The imaging payload draws 17.4 W for the Mast-Z camera during active imaging and 2 W in standby; the Innoflight CFC-400 flight computer dedicated to imaging consumes 20 W active and 3 W in standby. The refueling payload consists of four active-mode components: propellant line heaters at 30 W, valve actuators at 20 W, flow and pressure sensors at 10 W, and the control electronics board at 25 W; the refueling system draws 10 W in standby across all other modes. The propulsion subsystem is dominated by the SPT-100 Hall thruster at 1,350 W, with the Power Processing Unit (PPU) adding 135 W, estimated as 10% of thruster input power to account for the conversion losses between bus voltage and the thruster operating voltage. The ADCS subsystem includes four reaction wheel motors at a combined 213.7 W, 36 RCS solenoid valves at 144 W, 24 RCS flow meters at 96 W, 12 RCS pumps at 406.1 W, four gyroscopes at 10 W, 12 star trackers at 24 W, 12 sun sensors at 6 W, and four magnetometers at 2 W. The telecom subsystem contributes the Innoflight SCR-106 X-band transmitter at 0.1 W, the EnduroSat X-band patch antenna at negligible draw, and the TT&C S-band receiver at 3.0 W, which remains on in every mode so commands can be received at any time. The C&DH subsystem runs the Innoflight CFC-400 flight computer at 6.3 W in nominal operation and 2.0 W in safe or standby mode. The thermal subsystem includes survival heaters at 511.1 W, active only during eclipse, and thermal controllers and sensors at 10 W across all modes. The EPS itself contributes the PCDU at 193.5 W in every mode.

Subsystem	Component	CBE (W)	Cont.	MEV (W)
Imaging	Mast-Z Camera (active)	17.4	10%	19.1
	Mast-Z Camera (standby)	2.0	10%	2.2
	CFC-400 Flight Computer (active)	20.0	10%	22.0
	CFC-400 Flight Computer (standby)	3.0	10%	3.3
Refueling	Propellant Line Heaters	30.0	10%	33.0
	Valve Actuators	20.0	10%	22.0
	Flow and Pressure Sensors	10.0	10%	11.0
	Control Electronics Board	25.0	30%	32.5
	Refueling System (standby)	10.0	10%	11.0
Propulsion	SPT-100 Hall Thruster	1,350.0	10%	1,485.0
	Power Processing Unit (PPU)	135.0	10%	148.5
ADCS	Reaction Wheel Motors x4	213.7	10%	235.1
	RCS Solenoid Valves x36	144.0	10%	158.4
	RCS Flow Meters x24	96.0	10%	105.6
	RCS Pumps x12	406.1	10%	446.7
	Gyroscopes x4	10.0	10%	11.0
	Star Trackers x12	24.0	10%	26.4
	Sun Sensors x12	6.0	10%	6.6
	Magnetometers x4	2.0	10%	2.2
Telecom	SCR-106 X-band Transmitter	0.1	10%	0.1
	EnduroSat X-band Patch Antenna	0.0	10%	0.0
	TT&C S-band Receiver	3.0	10%	3.3
C&DH	Flight Computer - Nominal	6.3	20%	7.6
	Flight Computer - Safe/Standby	2.0	20%	2.4
Thermal	Survival Heaters - Eclipse	511.1	30%	664.4
	Thermal Controllers and Sensors	10.0	10%	11.0
EPS	PCDU	193.5	10%	212.9

Table 9-1: Power Breakdown - MEV

Contingency Margins

Contingency margins are applied per component based on design maturity. Components with flight-proven heritage carry a 10% contingency. Components with heritage designs that have been modified carry a 20% contingency. New or custom designs carry a 30% contingency. The control electronics board for the refueling system and the survival heaters carry 30% contingency because they involve non-standard configurations or custom software integration. The C&DH flight computer carries 20% because the software stack is adapted from heritage. Applying these margins to the Current Best Estimate (CBE) for each component produces the Maximum Estimated Value (MEV). A further 30% system margin is applied on top of the MEV totals for each mode before passing the demands to the EPS sizing analysis.

Operational Modes

Nine operational modes are defined, each corresponding to a distinct phase of the mission. The power demand per mode is shown in the tables below:

Standby mode occurs during cruise between targets, when no active operations are underway. Only housekeeping systems remain on: attitude control via reaction wheels, the TT&C receiver so ground commands can always be received, thermal monitoring, and the PCDU. All payloads drop to their minimum idle draw.

Nominal mode is the baseline operating state during orbital phases that require full situational awareness but no active payload operation. It adds the star trackers, sun sensors, and the main flight computer to the standby configuration, giving the spacecraft full attitude knowledge and command-handling capability.

Thrusting mode covers the Hall thruster burns used for the launch raise from 300 km to 550 km, the phasing hops between Starlink targets, and the end-of-life deorbit burn. The SPT-100 and its Power Processing Unit together account for the bulk of the demand, making this the highest-power mode of the mission and the one that sizes the solar array.

Rendezvous mode is entered during proximity operations, when the chaser closes the final distance to a target Starlink. The Hall thruster is off; the RCS takes over for fine trajectory and attitude control. All attitude sensors remain active because precise relative navigation is critical at close range.

Imaging mode is used to survey the target satellite before docking. The Mast-Z camera and its dedicated CFC-400 flight computer power up fully, and the RCS stays active to hold the chaser steady at the required standoff distance.

Refueling mode covers the docking and propellant transfer operation. The argon transfer hardware powers up: line heaters keep the propellant above its minimum operating temperature, valve actuators control the flow, pressure and flow sensors monitor the transfer, and the control electronics board manages the sequence. The RCS remains active throughout to maintain docking alignment.

Downlink mode is a short, periodic mode in which the X-band transmitter and patch antenna activate to relay stored data to the ground via NASA's DSN. All other systems remain at their nominal housekeeping levels.

Eclipse mode occurs during the roughly 35.5 minutes per orbit that the spacecraft spends in Earth's shadow. The solar array produces no power, so the battery takes over. Survival heaters activate to keep the spacecraft within its thermal operating limits, which is the dominant load during this mode and the one that sizes the battery.

Safe mode is a fault-response configuration. If the onboard computer detects an anomaly it cannot resolve autonomously, the spacecraft sheds all non-essential loads and enters a minimum-power state: only the reaction wheels, gyroscopes, magnetometers, TT&C receiver, thermal sensors, and the low-power flight computer remain on. This guarantees that attitude control and ground communication are preserved while the fault is diagnosed, at the lowest possible power draw of any mission phase.

Table 9-2: Operational Modes Tabulated

Component	Standby	Nominal	Image	Refuel	Thrust	Rendezvous	Downlink	Eclipse	Safe
Mast-Z Camera (active)	Off	On	On	Off	Off	Off	Off	Off	Off
Mast-Z Camera (standby)	On	On	Off	On	On	On	On	On	Off
CFC-400 (active)	Off	Off	On	Off	Off	Off	Off	Off	Off
CFC-400 (standby)	On	On	Off	On	On	On	On	On	Off
Refueling payload (active)	Off	Off	Off	On	Off	Off	Off	Off	Off
Refueling system (standby)	On	On	On	Off	On	On	On	On	Off
SPT-100 + PPU	Off	Off	Off	Off	On	Off	Off	Off	Off
Reaction Wheels x4	On	On	On	On	On	On	On	On	On
RCS (valves/meters/pumps)	Off	Off	On	On	On	On	Off	Off	Off
Gyroscopes x4	On	On	On	On	On	On	On	On	On
Star Trackers x12	Off	On	On	On	On	On	On	Off	Off
Sun Sensors x12	Off	On	On	On	On	On	On	Off	Off
Magnetometer x4	On	On	On	On	On	On	On	On	On
X-band Transmitter	Off	Off	Off	Off	Off	Off	On	Off	Off
TT&C Receiver (S-band)	On	On	On	On	On	On	On	On	On
Flight Computer - Nominal	Off	On	On	On	On	On	On	On	Off
Flight Computer - Safe/Stby	On	Off	Off	Off	Off	Off	Off	Off	On
Survival Heaters	Off	Off	Off	Off	Off	Off	Off	On	Off
Thermal Controllers/Sensors	On	On	On	On	On	On	On	On	On
PCDU	On	On	On	On	On	On	On	On	On

Table 9-3: Final MEV Breakdown with Margin

Mode	CBE (W)	MEV (W)	MEV + 30% Margin (W)
Standby	449.2	494.3	642.6
Nominal	483.5	532.5	692.2
Imaging	1,162.0	1,278.8	1,662.5
Refueling	1,204.6	1,330.7	1,729.9
Thrusting	2,614.6	2,876.7	3,739.7
Rendezvous	1,129.6	1,243.2	1,616.2
Downlink	483.6	532.6	692.4
Eclipse	964.6	1,163.9	1,513.0
Safe Mode	434.2	477.8	621.2

After applying the 30% system margin, mode demands range from 621.2 W in safe mode to 3,739.7 W in thrusting mode. Thrusting mode produces the highest power demand and drives the solar array size; eclipse mode drives the battery capacity. The PCDU losses (193.5 W, present in every mode) are subtracted from the mode totals before they enter the array sizing calculation, since those losses are already captured in the path efficiencies used in that calculation. Double-counting them would result in an oversized array.

10 - Electrical Power System

The Electrical Power System (EPS) provides, stores, and distributes power to all spacecraft subsystems. It consists of three elements: a deployable solar array that generates power during sunlight, a lithium-ion battery that sustains the spacecraft during eclipse, and a Power Control and Distribution Unit (PCDU) that regulates the bus voltage and routes power to each load. Two modes from the PEL drive the sizing: thrusting mode sets the solar array requirement and eclipse mode sets the battery requirement. The power values used as inputs to the array sizing are not taken directly from the PEL mode totals. The PCDU losses (193.5 W) are already captured in the path efficiencies of the array sizing equation, explained in detail in the Solar Array section below; including them again as a load would double-count them. The sizing inputs are therefore the PEL MEV totals minus the PCDU loss contribution, with the 30% system margin applied: 3,463 W for thrusting mode and 1,236 W for eclipse mode.

Orbital Context

The EPS derives its sizing inputs from the mission orbit defined by the trajectory subsystem: 550 km circular at 53° inclination. For power sizing, a worst-case beta angle of 0° is assumed, since this maximises eclipse duration and produces the most conservative array and battery requirements. Under these conditions the orbital period is 95.5 minutes, with a 35.6-minute eclipse per orbit (37.2% eclipse fraction). The actual operational mission lasts approximately one year per chaser (as derived in the trajectory analysis), but the project requirements mandate a 5-year design life for the power system. The EPS is therefore sized for 5 years, not for the expected mission duration. This is standard practice in spacecraft design: hardware is qualified and sized to a design life that exceeds the operational mission to provide margin against schedule delays, extended operations, and potential mission reuse. Over 5 years the spacecraft completes 27,518 eclipse cycles, and this cycle count is the primary constraint on battery cell selection.

Solar Array

The solar array uses **Azur Space 3G30C-Advanced** triple-junction GaAs cells, selected for their flight-proven performance in LEO missions and their well-characterised radiation degradation data. The beginning-of-life (BOL) cell efficiency is 29.5%, taken directly from the Azur Space datasheet (Typical Electrical Performance table, η at 1,367 W/m², BOL column). Cell and coverglass mass (1.76 kg/m²) is taken from the Spectrolab heritage datasheet, which uses the same GaAs triple-junction technology [12].

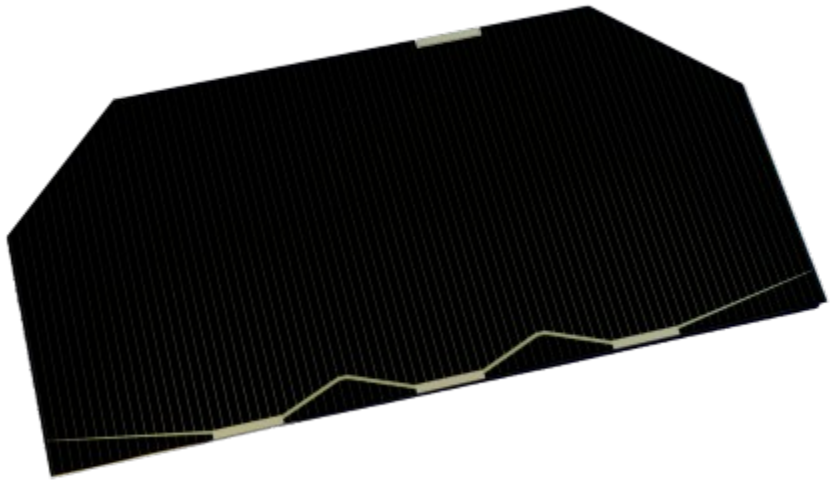


Figure 10.1: Azur Space 3G30C-Advanced triple-junction GaAs solar cell

The end-of-life (EOL) efficiency factor accounts for radiation-induced degradation over the 5-year mission. The highest fluence column applicable to LEO in the Azur Space datasheet is 1×10^{15} e^{-}/cm^2 (1 MeV electrons), giving $\eta = 27.1\%$ at EOL. The EOL factor is 27.1 / 29.5 = 0.919. Using the most conservative column ensures the array is not undersized. The BOL output per unit area, with a panel packing factor of 0.90 (90% of panel area covered by cells, consistent with NASA practice per Wertz & Larson, *Spacecraft Electrical Power Systems*), is:

$$P_BOL/m^2 = S \times \eta_cell \times f_pack = 1{,}360 \times 0.295 \times 0.90 = 361.1\ W/m^2$$

The EOL output per unit area is:

$$P_EOL/m^2 = P_BOL/m^2 \times f_EOL = 361.1 \times 0.919 = 331.8\ W/m^2$$

With the BOL and EOL outputs per unit area established, the remaining question is how much total power the array needs to produce. The answer is not simply the peak load from the PEL, because the array has to do two jobs at once: power the spacecraft during sunlight and, simultaneously, recharge the battery so it can sustain the spacecraft through the next eclipse. To find the array output that satisfies both constraints together, the sizing uses the power balance equation from Wertz & Larson, *SMAD*, Ch. 11, a standard reference in spacecraft power system design[13]. The equation expresses the required array power P_sa as a function of four inputs: the power demand during eclipse (P_e), the power demand during sunlight (P_d), the duration of each phase per orbit (t_ecl and t_sun), and two path efficiencies that capture the losses between the array and the loads. The sunlight path efficiency $X_e = 0.85$ covers losses from the solar array output through the PCDU to the loads, and is the upper end of the typical range for LEO deployable GaAs arrays per Wertz & Larson (*Spacecraft Electrical Power Systems*, Spacecraft Electrical Power Systems). The eclipse path efficiency $X_n = 0.96$ covers losses from the battery through the PCDU to the loads, set equal to the PCDU conversion efficiency; battery discharge losses are covered separately by the 15% safety margin applied in the battery sizing.

$$P_sa = [\ (P_e \times t_ecl / X_n) + (P_d \times t_sun / X_e)\ ] / t_sun$$

Where P_e = 1,236 W, t_ecl = 2,133 s, X_n = 0.96, P_d = 3,463 W, t_sun = 3,597 s, X_e = 0.85:

$$P_sa = [\ (1{,}236 \times 2{,}133 / 0.96) + (3{,}463 \times 3{,}597 / 0.85)\ ] / 3{,}597 = 4{,}837.7\ W$$

The minimum array area is:

$$A_min = P_sa / P_EOL/m^2 = 4{,}837.7 / 331.8 = 14.58\ m^2$$

The array splits into two wings of 7.29 m² each. At BOL the array produces 5,264 W; at EOL it produces 4,838 W, matching the required output with 0% residual margin by design, since the 30% system margin is already embedded in the PEL mode demands.

Table 10-1: Solar Array Properties

Parameter	Value
Cell type	Azur Space 3G30C-Advanced
BOL cell efficiency	29.5%
EOL efficiency factor	0.919
Packing factor	0.90
BOL output per m²	361.1 W/m²
EOL output per m²	331.8 W/m²
Required array power (P_sa)	4,837.7 W
Total array area	14.58 m²
Area per wing	7.29 m²
Array BOL power	5,264 W
Array EOL power	4,838 W
Cells + coverglass mass	25.7 kg
Wing structure + hinges mass	21.9 kg
Deployment mechanism mass	10.0 kg
TOTAL SOLAR ARRAY MASS	57.5 kg

Battery

The battery uses **Saft VES16** lithium-ion cells. The VES16 is selected for its demonstrated cycle life in LEO: Saft's Space Power Workshop 2019 data shows less than 12% energy loss after 12 years at 30% depth of discharge, and a rated cycle life above 60,000 cycles. Sized to the 5-year design life, the mission accumulates 27,518 eclipse cycles, which the VES16 covers with a 2.18× margin [9].



The battery is sized to sustain the spacecraft through one worst-case eclipse. The eclipse energy demand is:

$$E_{ecl} = P_e \times t_{ecl} = 1{,}236 \times (2{,}133 / 3{,}600) = 732.5 \text{ Wh}$$

The depth of discharge (DoD) is set at 30%, consistent with the Saft VES16 recommended range of 20–40% for LEO. Dividing by DoD gives the gross capacity needed:

$$C_{gross} = E_{ecl} / DoD = 732.5 / 0.70 = 1{,}046.4 \text{ Wh}$$

The battery loses capacity over its 5-year life. From Saft SPW 2019, the 12-year fade at 30% DoD is 12%; linearly interpolating to 5 years gives a 5% loss, so the EOL factor is 0.95:

$$C_{BOL} = C_{gross} / f_{EOL} = 1{,}046.4 / 0.95 = 1{,}101.5 \text{ Wh}$$

A 15% safety margin covers modelling uncertainties:

$$C_{design} = C_{BOL} \times 1.15 = 1{,}101.5 \times 1.15 = 1{,}266.7 \text{ Wh}$$

Battery mass uses the minimum specific energy of 155 Wh/kg from the VES16 datasheet:

$$m_{bat} = C_{design} / e_{spec} = 1{,}266.7 / 155 = 8.2 \text{ kg}$$

The cycle life check confirms the VES16 survives the full mission with margin:

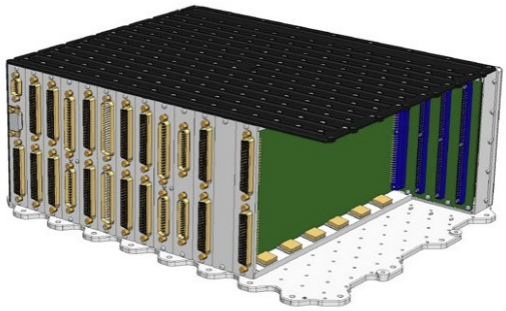
$$\text{Cycle margin} = 60{,}000 / 27{,}518 = 2.18$$

Table 10-2: Battery Properties

Parameter	Value
Cell type	Saft VES16
Depth of discharge	30%
Eclipse energy demand	732.5 Wh
Gross capacity (÷ DoD)	1,046.4 Wh
EOL capacity factor	0.95
Required BOL capacity	1,101.5 Wh
Design capacity (+15% margin)	1,266.7 Wh
Specific energy	155 Wh/kg
Battery mass	8.2 kg
Mission eclipse cycles	27,518
VES16 rated cycle life	>60,000
Cycle life margin	2.18×

Power Control and Distribution Unit (PCDU)

The PCDU uses a **Terma PCDU** with Sequential Switching Shunt Regulator (S3R) topology. The S3R architecture regulates bus voltage by shunting excess solar array current rather than dissipating it as heat, making it the most efficient topology for LEO solar-powered spacecraft. The PCDU efficiency is estimated at 96%, based on a 2021 MDPI Energies paper on S3R-based power bus management[10]. Terma does not publish an exact efficiency value, but their PCDU uses the same S3R architecture, so 96% is the appropriate estimate[11].



The bus voltage is set at 100 V, the maximum supported by the Terma PCDU[11]. A higher bus voltage reduces bus current for a given power level, minimising resistive losses in the harness. At EOL the bus current is:

$$I_{bus} = P_{EOL} / V_{bus} = 4{,}838 / 100 = 48.4\ A$$

PCDU losses at EOL are:

$$P_{loss} = P_{EOL} \times (1 - \eta_{PCDU}) = 4{,}838 \times 0.04 = 193.5\ W$$

These losses appear as a continuous thermal load and are accounted for in the PEL as a fixed draw across all modes. They are excluded from the SMAD sizing input to avoid double-counting, since they are already captured in the path efficiencies X_e and X_n. PCDU and harness mass is estimated at 20 kg (5 kg for the PCDU unit, 15 kg for harness), consistent with small satellite heritage at this power level.

Table 10-3: PCDU Properties

Parameter	Value
PCDU type	Terma S3R
Bus voltage	100 V
PCDU efficiency	96%
Bus current (EOL)	48.4 A
PCDU losses	193.5 W
PCDU + harness mass	20.0 kg

EPS Mass and Power Summary

The three EPS subsystems together supply all power needs across the nine operational modes defined in the PEL. The solar array provides 0% residual margin over the required 4,838 W at EOL by design, since the 30% system margin is already embedded in the PEL mode demands. The battery cycle margin of 2.18× over the mission requirement gives confidence in the VES16 selection for the 5-year design life.

Table 10-4: Mass Breakdown

Subsystem	Component	Mass (kg)
Solar Array	Cells + coverglass	25.7
	Wing structure + hinges	21.9
	Deployment mechanisms	10.0
Battery	Saft VES16 pack	8.2
PCDU	PCDU unit + harness	20.0
TOTAL EPS MASS		85.7

References (cited in these sections)

[9] Borthomieu, Y., Armel, V., Le Guern, B., & Ma, C. (2019, April 2). VES16 batteries for LEO and GEO satellites [Conference presentation]. Space Power Workshop, Torrance, CA. https://spw.aerospace.org/files/2021/07/2019_04_02_IV-e_Borthomieu.pdf

[10] Schirone, L., Ferrara, M., Granello, P., Paris, C., & Pellitteri, F. (2021). Power bus management techniques for space missions in low Earth orbit. Energies, 14(23), 7932. https://doi.org/10.3390/en14237932

[11] Terma A/S. (2025). Power conditioning and distribution units: For all types of satellites [Brochure]. https://www.terma.com/media/fpfjapsa/brochure-terma-space-power-conditioning-and-distribution-units-a4.pdf

[12] AZUR SPACE Solar Power GmbH. (n.d.). TJ3G30-Advanced 4 x 8 solar cell [Datasheet]. https://www.azurspace.com/media/uploads/file_links/file/bdb_00010891-01-00_tj3g30-advanced_4x8.pdf

[13] Larson, W. J., & Wertz, J. R. (1999). Space mission analysis and design (3rd ed.). Microcosm Press.